

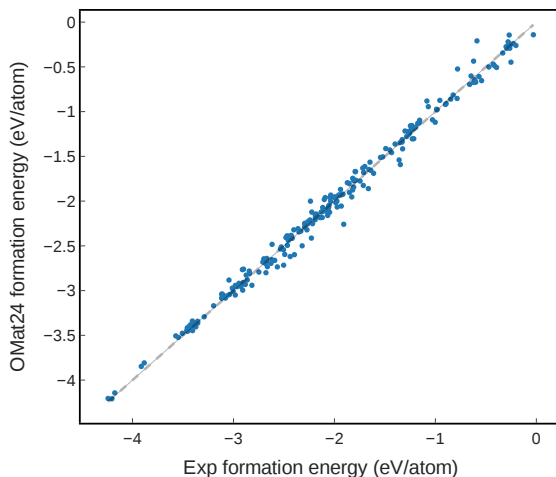


Figure C.1 OMat24 DFT vs experimental formation energy parity. Formation energy calculated with OMat24 DFT settings vs experimental formation energy.

reiterate that the OMat24 references should be used when computing formation energies with OMat24 trained models. Similarly, we recommend that the OMat24 compatibility corrections be used when computing formation energies involving compounds for which a +U correction was or would have been used in the DFT calculation accordingly. Furthermore, we remind readers that when using MP2020 style compatibility corrections, the predicted formation energies are meant to represent values at room temperature (the temperature of experimental formation energies used to fit the corrections). We refer readers to the original publication⁴¹ detailing the correction scheme to get a full understanding of the details and nuances involved in these calculations.

D Dataset Learning Curves

Figure D.1 presents the energy and force mean absolute error (MAE) learning curves for OMat24 training subsamples of increasing size using the eSEN architecture. Consistent with general machine learning principles, we observe that prediction errors decrease as the dataset size increases, provided the model has sufficient capacity. Notably, for the eSEN architecture in these experiments, the error plateaus at approximately 100 million samples, which matches the total size of the OMat24 training set.

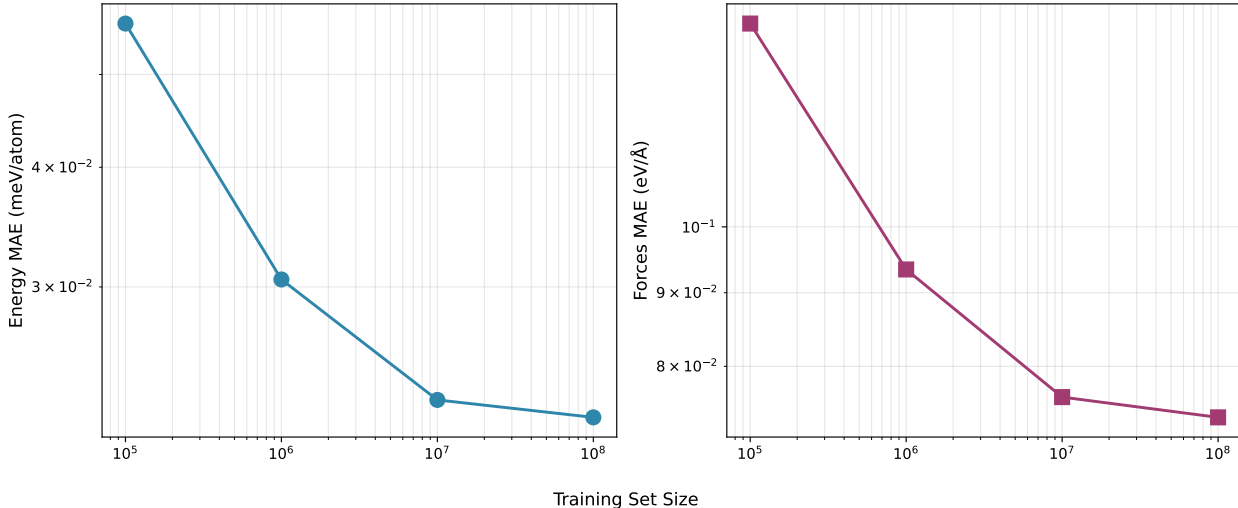


Figure D.1 Learning curves for energy and force predictions of the eSEN model on OMat24 subsamples. Energy mean absolute error (MAE) and forces MAE as a function of number of structures in training set.

E Systematic softening improvements across model architectures

Figure E.1 presents violin plots illustrating the various orders of softening, as introduced in the main text, for two additional architectures: GRACE and Sevensn. Improvements across all degrees of softening are consistently observed in five distinct architectures, indicating that the effect is largely architecture-independent and primarily attributable to the diversity of the OMat24 dataset.

F Effects of denoising augmentation versus dataset diversity

We investigate whether denoising augmentation can produce accurate models with smaller datasets and further enhance accuracy when combined with OMat24.

To assess the impact of denoising augmentation and establish a performance baseline, we utilize the denoising nonequilibrium structures (DeNS) protocol²⁷. We compare models trained exclusively on the MPtraj dataset, both with and without DeNS, which are classified as *compliant* in the Matbench Discovery benchmark²⁹. This allows a structured comparison of architectures and the effects of DeNS on a fixed dataset. Subsequently, we also evaluate the impact of training on the OMat24 dataset, again with and without DeNS.

Table F.1 presents the results of the best compliant models on the *discovery* task of the Matbench Discovery benchmark. The eSEN and equiformerV2 models with DeNS outperform their non-DeNS counterparts and all other third-party models in this task. For the larger equiformerV2 model, DeNS regularization facilitates effective training even with the smaller MPtraj dataset. The energy MAE remains consistent (35-36 meV/atom) across different equiformerV2 sizes, indicating the practical utility of the smaller model. These findings suggest that dataset augmentation techniques like DeNS can significantly enhance performance when training solely on the MPtraj dataset.

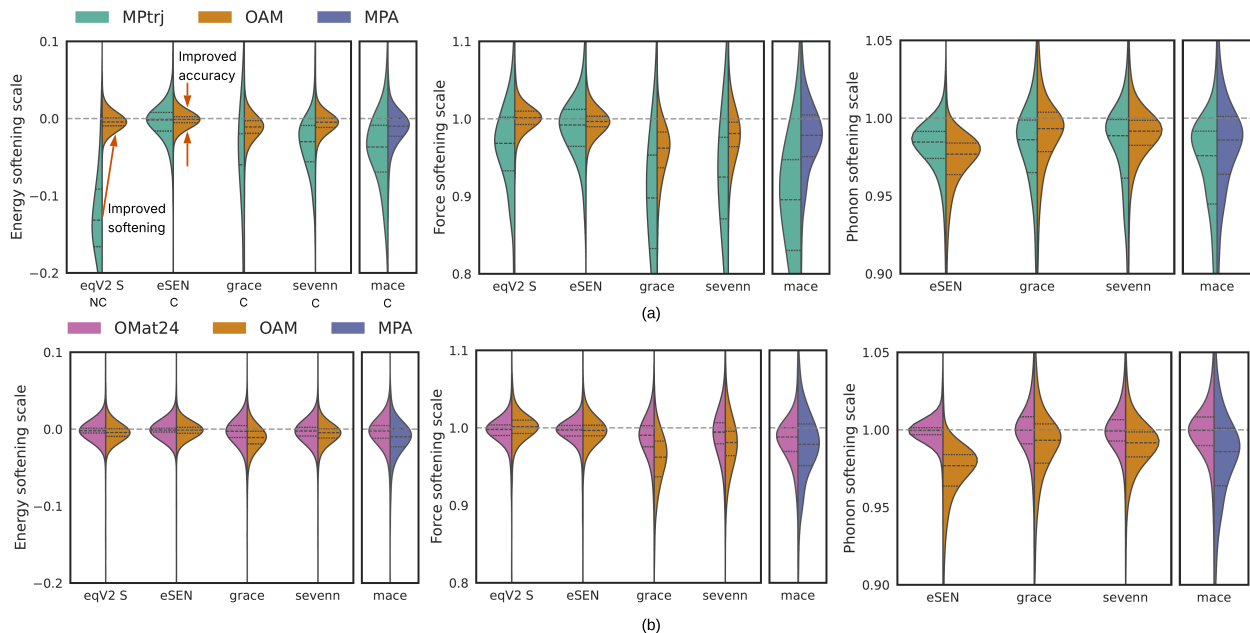


Figure E.1 Systematic softening improvements across additional model architectures. Zeroth order (energy), first order (forces) and second order (phonon) systematic softening distribution shifts between (a) MPTrj, OAM (OMat24 + MP/sAlex finetuning) and MPA (MP/sAlex) trained models and (b) OMat24, OAM, and MPA models. Violin plots show the distribution of the energy and force softening scale of model predictions on high energy WBM set²³, and phonon softening of conservative model predictions on the recalculated MDR phonon dataset⁴⁹. Models are labeled (C) *conservative* if forces are predicted as energy gradients, and (NC) *not conservative* otherwise.

Table F.1 Matbench-Discovery benchmark results of compliant models. benchmark results of compliant models trained only on MPtrj with results on the unique prototype split. Mean absolute error (MAE) and Root mean squared error (RMSE) are in units of eV/atom.. Top third party models from Matbench-Discovery snapshot before May 1, 2025.

Model	eSEN	eqV2-L-DeNS	eqV2-M-DeNS	eqV2-S-DeNS	eqV2-S	MatRIS v0.5.0	ORB v2	SevenNet
F1 ↑	0.831	0.823	0.818	0.815	0.77	0.809	0.765	0.724
DAF ↑	5.260	5.184	5.109	5.042	4.64	5.049	4.702	4.252
Precision ↑	0.804	0.792	0.781	0.771	0.709	0.772	0.719	0.65
Recall ↑	0.861	0.856	0.858	0.864	0.841	0.850	0.817	0.818
Accuracy ↑	0.946	0.944	0.942	0.941	0.926	0.938	0.922	0.904
TPR ↑	0.861	0.856	0.858	0.864	0.841	0.850	0.817	0.818
FPR ↓	0.038	0.041	0.044	0.047	0.063	0.46	0.059	0.081
TNR ↑	0.962	0.959	0.956	0.953	0.937	0.954	0.941	0.919
FNR ↓	0.139	0.144	0.142	0.136	0.159	0.150	0.183	0.182
MAE ↓	33	35	35	36	42	37	45	48
RMSE ↓	78	82	82	85	87	82	91	92
R2 ↑	0.822	0.802	0.803	0.788	0.778	0.803	0.756	0.75

Figure F.1 (a) depicts the rolling mean absolute error of energy above hull predictions using a 40 meV/atom window for small equiformerV2 models trained on various datasets. Non-zero errors within the gray triangle indicate that the model error exceeds the true energy above hull, and with probability leading to incorrect stability classifications.

We observe an overall improvement when training with DeNS on MPtraj compared to using MPtraj alone. More pronounced performance gains are evident in models pre-trained on larger, diverse datasets. Pre-training on OC20, a diverse catalysis dataset, slightly enhances performance over DeNS after fine-tuning on MPtraj for structures near the convex hull. Furthermore, models trained on OMat24, an in-domain dataset, exhibit the most significant performance improvements. Training on OMat24 results in models with a broader window of

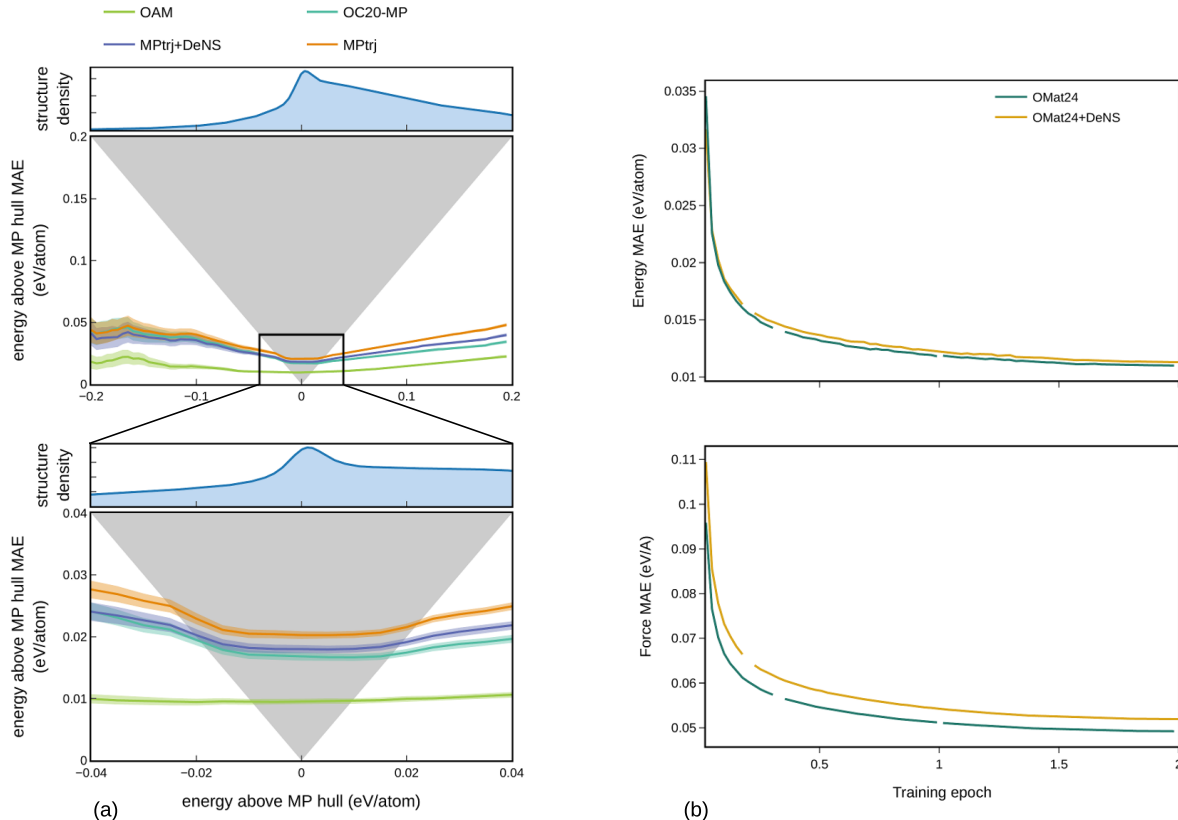


Figure F.1 Impact of denoising augmentation and dataset diversity on model accuracy and validation errors. (a) Rolling mean absolute error of predicted energy above hull for equiformerV2 small models (31M parameters) using a window of 40 meV/atom. (b) energy and force mean absolute error validation curves during equiformerV2 small models training on OMat24 with DeNS and without it.

constant accuracy, indicating better extrapolation to regions with lower dataset density farther away from the convex hull.

Finally, Figure F.1 (b) presents the energy and force validation mean absolute error during the training of equiformerV2-S models on OMat24, both with and without DeNS. We observe a slight performance penalty when using DeNS with OMat24, indicating that the diversity of the larger OMat24 dataset does not benefit from the regularization or augmentation provided by DeNS.

G Model training hyper-parameters and configuration

The models trained using the EquiformerV2 architecture²⁶ have separate heads for energy, forces and stress prediction. We use a per-atom MAE loss for energy and an l_2 norm loss for forces. For stress prediction, we decompose the 3×3 symmetric stress tensor to a 1-dimensional isotropic component ($L = 0$) and a 5-dimensional anisotropic component ($L = 2$). We use a prediction head that outputs a scalar and irreps of $L = 2$, then use an MAE loss for the isotropic and the anisotropic component separately. At test time, we recover the stress tensor by combining the isotropic and anisotropic components. For models trained with DeNS, an additional head is added to predict the noise added to a perturbed structure. For a DeNS forward pass, we input the noisy structure and predict the unperturbed energy as well as the noise vectors, then compute a per-atom MAE loss for energy and an MSE loss for the noise vectors. We refer interested readers to the original EquiformerV2 and DeNS papers for more details on model architectures^{26,27}. The hyper-parameters for our models are summarized in Table G.1. The hyper-parameters for training are